

A Practical Guide to Principal component analysis

TOPIC -- PRINCIPAL COMPONENT ANALYSIS

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where the eigenvalue property of $w(k)$ has been used to move from line 2 to line 3. However eigenvectors $w(j)$ and $w(k)$ corresponding to eigenvalues of a symmetric matrix are orthogonal (if the eigenvalues are different), or can be orthogonalised (if the vectors happen to share an equal repeated value). The product in the final line is therefore zero; there is no sample covariance between different principal components over the dataset. Another way to characterise the principal components transformation is therefore as the transformation to coordinates which diagonalise the empirical sample covariance matrix. In matrix form, the empirical covariance matrix for the original variables can be written

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The quantity to be maximised can be recognised as a Rayleigh quotient. A standard result for a positive semidefinite matrix such as XTX is that the quotient's maximum possible value is the largest eigenvalue of the matrix, which occurs when w is the corresponding eigenvector. With $w(1)$ found, the first principal component of a data vector $x(i)$ can then be given as a score $t1(i) = x(i) \cdot w(1)$ in the transformed co-ordinates, or as the corresponding vector in the original variables, $\{x(i) \cdot w(1)\} w(1)$.

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$$\mathbf{q}_i = \mu + z_i, \mu = \frac{1}{n} \sum_{i=1}^n q_i. \quad \mathbf{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{q}_i.$$

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Write $x_1 \dots x_n$ as row vectors, each with p elements. Place the row vectors into a single matrix X of dimensions $n \times p$.

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A has full column rank Each column of A must have at least $L - 1$ zeroes where L is the number of columns of A (or alternatively the number of rows of

P). The justification for this criterion is that if a node is removed from the regulatory layer along with all the output nodes connected to it, the result must still be characterized by a connectivity matrix with full column rank.

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Matrix V , also of dimension $p \times p$, contains p column vectors, each of length p , which represent the p eigenvectors of the covariance matrix C . The eigenvalues and eigenvectors are ordered and paired. The j th eigenvalue corresponds to the j th eigenvector. Matrix V denotes the matrix of right eigenvectors (as opposed to left eigenvectors). In general, the matrix of right eigenvectors need not be the (conjugate) transpose of the matrix of left eigenvectors.

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Use the vector g as a guide in choosing an appropriate value for L . The goal is to choose a value of L as small as possible while achieving a reasonably high value of g on a percentage basis. For example, you may want to choose L so that the cumulative energy g is above a certain threshold, like 90 percent. In this case, choose the smallest value of L such that $\frac{g_L}{g_p} \geq 0.9$

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$$\begin{aligned} X^T X &= W \Sigma^T U^T U \Sigma W^T = W \Sigma^T \Sigma W^T = W \Sigma^2 W^T \\ \begin{aligned} \mathbf{X}^T \mathbf{X} &= \mathbf{W} \mathbf{\Sigma}^T \mathbf{U}^T \mathbf{U} \mathbf{\Sigma} \mathbf{W}^T = \mathbf{W} \mathbf{\Sigma}^T \mathbf{\Sigma} \mathbf{W}^T = \mathbf{W} \mathbf{\Sigma}^2 \mathbf{W}^T \\ \mathbf{X}^T \mathbf{X} &= \mathbf{W} \mathbf{\Sigma}^T \mathbf{U}^T \mathbf{U} \mathbf{\Sigma} \mathbf{W}^T = \mathbf{W} \mathbf{\Sigma}^T \mathbf{\Sigma} \mathbf{W}^T = \mathbf{W} \mathbf{\Sigma}^2 \mathbf{W}^T \\ \mathbf{X}^T \mathbf{X} &= \mathbf{W} \mathbf{\Sigma}^T \mathbf{U}^T \mathbf{U} \mathbf{\Sigma} \mathbf{W}^T = \mathbf{W} \mathbf{\Sigma}^T \mathbf{\Sigma} \mathbf{W}^T = \mathbf{W} \mathbf{\Sigma}^2 \mathbf{W}^T \end{aligned} \end{aligned}$$

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Computation using the covariance method The following is a detailed description of PCA using the covariance method as opposed to the correlation method. The goal is to transform a given data set X of dimension p to an alternative data set Y of smaller dimension L . Equivalently, we are seeking to find the matrix Y , where Y is the Karhunen–Loève transform (KLT) of matrix X :

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Online/sequential estimation In an "online" or "streaming" situation with data arriving piece by piece rather than being stored in a single batch, it is useful to make an estimate of the PCA projection that can be updated sequentially. This can be done efficiently, but requires different algorithms.

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where y is a q -element vector and B' is a $(q \times p)$ matrix, and let $\Sigma_y = B' \Sigma B$ be the variance-covariance matrix for y . Then the trace of Σ_y , denoted $\text{tr}(\Sigma_y)$, is maximized by taking $B = A_q$, where A_q consists of the first q columns of A (B' is the transpose of B). (A is not defined here) Property 2: Consider again the orthonormal transformation

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Directional component analysis Directional component analysis (DCA) is a method used in the atmospheric sciences for analysing multivariate datasets. Like PCA, it allows for dimension reduction, improved visualization and improved interpretability of large data-sets. Also like PCA, it is based on a covariance matrix derived from the input dataset. The difference between PCA and DCA is that DCA additionally requires the input of a vector direction, referred to as the impact. Whereas PCA maximises explained variance, DCA maximises probability density given impact. The motivation for DCA is to find components of a multivariate dataset that are both likely (measured using probability density) and important (measured using the impact). DCA has been used to find the most likely and most serious heat-wave patterns in weather prediction ensembles, and the most likely and most impactful changes in rainfall due to climate change.

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$w_{(1)} = \arg \max_{\|w\|=1} \left\{ \sum_i (t_{1i})^2 \right\} = \arg \max_{\|w\|=1} \left\{ \sum_i (x_i \cdot w)^2 \right\}$